

Personalised AI Assisted Masters Programme Selection

Business and Technical Trade off Analysis | NIU DUOER | PE6201 Section B | 30 September 2026

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Problem and scope

I built DS Compass for international graduates who need an initial, defensible way to compare master's programmes. The primary user is an applicant who has a broad study or career goal but must navigate different programme names, entry rules and English-language requirements across university pages. The closest existing alternative is manual browsing of official university pages. It is authoritative, but the applicant must translate their goal into search terms and compare requirements across multiple pages. My prototype reduces that first research step to a source-linked shortlist. It does not predict admission, score universities, calculate an applicant's chance, submit applications, or make an academic-equivalence decision.

The original proposal described a foundation model plus official-source retrieval and anonymised historical cases. I retained the foundation-model and official-source elements, but did not use the supplied historical case document. It has no reliable programme-level labels, contains personal application records, and its reuse permission and date range are not confirmed. Publishing it or treating it as admissions ground truth would create both privacy and validity problems. I therefore expanded the catalogue from the initial Singapore proof of concept to 15 official programme records across Singapore and the United Kingdom. This fits the proposal's original cross-country comparison scope while remaining small enough to verify.

Design and trade offs

The end-to-end path is profile entry, a selected study direction or short keyword classification, same-domain programme retrieval, English-evidence checking, and display of official links. The system rents OpenRouter's `google/gemini-2.5-flash-lite` only when a user selects Other or unsure and enters short keywords. The model must return one of seven labels: AI, data, systems, computing conversion, business, education, or abstain. It cannot generate programme facts, eligibility decisions, or admissions predictions. A local catalogue then retrieves only cards in the selected country and classified domain. This is lightweight retrieval over a small verified dataset, not a claim of a large-scale RAG system.

I chose an LLM because Chinese and English applicants often express their interests in varied language, while a narrow fixed schema controls the output. I retained a Laplace-smoothed Naive Bayes English baseline. If the model service is unavailable, the interface identifies the fallback rather than silently substituting it. The baseline is cheap and reproducible but cannot support Chinese text well. The LLM has per-call cost and outage risk, so the service key stays server-side, inputs are length-limited, and requests are rate-limited. The model is not given historical cases or official-page text; this reduces privacy exposure and unsupported factual generation.

I encoded IELTS minimums and component-score rules where the selected university page supported them. The form separates IELTS Listening, Reading, Writing and Speaking. It also records TOEFL iBT but never converts it into IELTS because the small catalogue does not encode a comparable TOEFL rule for every programme. The interface assumes a next-autumn plan and asks the user to confirm test validity directly with each university. Academic background, GPA conversion, work experience and exemptions are likewise manual-review evidence, never automatic eligibility.

Data, evaluation and cost

The catalogue contains 15 author paraphrases of official programme pages, each with a source URL and a 30 September 2026 verification date. A 90-day source-age prompt asks the user to check again. The language model test set was generated by GPT-4.1 mini before the Gemini evaluation. Although I requested 42 cases, the generator returned 37 valid labelled cases; I froze that actual output rather than adding cases after seeing results. The set covers the six in-domain labels and abstention prompts.

Gemini 2.5 Flash Lite achieved 86.5% exact-label accuracy on the 37 frozen synthetic cases. The majority-class baseline was 18.9%. The model abstained on 8.1% of cases; none of those abstentions would otherwise have been an error against the synthetic labels. These numbers measure primary-interest classification only. They do not measure programme quality, eligibility, user time saved or admission outcomes. The evaluation is small, synthetic and model-authored, so it is not independent human-expert validation. I report it as an engineering check, not a performance guarantee.

The independent dataset generation cost recorded by OpenRouter was approximately USD 0.0022. The deployed model's published pricing at the time of implementation was USD 0.10 per million input tokens and USD 0.40 per million output tokens; the short, capped classification request is therefore expected to cost far below one US cent, but live cost varies by provider and is not guaranteed. The hosting layer holds the secret and serves the worker; the user interface, retrieval catalogue, evaluation files, policy rules and source links are project-owned code and data.

Risks and acceptance test

The important risks are stale university requirements, an overconfident model label, a misleading English-evidence status, and user over-reliance. Mitigations are a source date, official links, fixed labels, abstention, a visible fallback label, no probability output, and manual-review text. A silent failure remains possible if a mixed goal receives a plausible single label. Another is an incorrectly transcribed university rule within the 90-day window. The next iteration should use independently human-labelled queries, a separate source reviewer and a development set for threshold calibration.

The acceptance test is concrete: an applicant selects Singapore and/or the United Kingdom, selects a study direction, optionally enters IELTS or TOEFL evidence, submits the form, and receives either a source-linked same-domain shortlist with an English-evidence status or a clear abstention. The deployed site completes this path. A final submission must also include the code repository and a narrated demonstration that states these limitations aloud.

Sources

University programme and English-language pages linked directly from each of the 15 DS Compass result cards. OpenRouter API reference and model pricing were checked on 30 September 2026.